

MARCELO AUGUSTO FRIES

Forward Deployed Engineer

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PROFESSIONAL SUMMARY

Forward Deployed Engineer who embeds with organizations that have no in-house engineering team and turns their operational problems into production systems, working solo from discovery to handover. Since 2024: scoping requirements with stakeholders, designing architectures, integrating third-party platforms (WhatsApp Business Cloud API, Google Android Management API, Mercado Pago Pix) and deploying to self-hosted and cloud infrastructure. Every system ships with runbooks and guides that non-technical staff use to run it. My work spans web, mobile, desktop, workflow automation and DevOps, mainly in TypeScript, Node.js, React/Next.js, PostgreSQL, Docker and Kotlin. Professional working proficiency in English; available during US and European business hours.

TECHNICAL SKILLS

FDE Practice: Customer Discovery & Requirements, Solution Architecture, Third-Party API Integration, Self-Hosted Deployment, Data-Privacy Compliance (LGPD), Production Debugging, Technical Documentation & Handover

Languages: TypeScript, JavaScript, SQL, Python, Kotlin, Bash, HTML5, CSS3

Backend & Data: Node.js, Fastify, REST APIs, Zod, PostgreSQL, TimescaleDB, Redis, BullMQ, Supabase (RLS, Edge Functions, Realtime, pg_cron), Neon

Frontend & Clients: React, Next.js (App Router), Astro, Vite, Tailwind CSS, Zustand, Android (Kotlin, Room, WorkManager), NW.js

Integrations & Automation: WhatsApp Business Cloud API, Chatwoot, n8n, Google Android Management API, Cloud Pub/Sub, Mercado Pago Pix, Webhooks, Steamworks

Infrastructure & DevOps: Docker Compose, Linux, Nginx, TLS/Let's Encrypt, Cloudflare, S3-Compatible Storage, GitHub Actions, CI/CD, Vercel, Backup & Restore Automation

Security: JWT with Refresh-Token Rotation, RBAC, Row-Level Security, HMAC/OIDC Webhook Verification, Rate Limiting, Audit Logging

AI-Assisted Engineering: Claude Code, Claude API, Spec-Driven Development

EXPERIENCE

Forward Deployed Engineer

Feb 2026 – Present

Pública Assessoria em Gestão Empresarial Ltda · Contract · Brasília, Brazil (Hybrid)

Sole engineer embedded with a consultancy delivering technology to public-sector clients that have no internal engineering team. Project details are generalized due to NDA.

Citizen-Service Messaging Platform · WhatsApp Cloud API · Chatwoot · n8n · PostgreSQL · Redis · Docker

- Consolidated dozens of fragmented contact channels into a single official WhatsApp Business number. Deployed a self-hosted service desk (Chatwoot, n8n, PostgreSQL, Redis, Nginx) with Docker Compose on a hardened Linux VM to meet data-residency requirements.
- Built a rules-based triage bot, a 10-state conversation state machine in n8n that routes citizens to the right team and hands them off to human agents in a shared inbox.
- Engineered a spreadsheet-to-WhatsApp outreach pipeline: upload → data normalization → contact creation → approved template sends. Parallelizing dispatch within API rate limits made it **~8x faster** (~1.5k → ~14k messages per 30 min). Also fixed the out-of-memory crashes and timeouts that large batches were causing in production.
- Handled messy real-world data (Excel date serials, inconsistent phone formats) with per-row error logging. One templated deploy script generates 11 workflows. Backfilled 2.8k past conversations into standardized labels.
- Implemented LGPD privacy flows (privacy notice, opt-out, deletion requests), key-only SSH and encrypted nightly off-site backups with a tested restore. Wrote operator and IT guides that let staff run the platform without me.

Android Device Management Platform · TypeScript · Fastify · Next.js · Kotlin · TimescaleDB · BullMQ

- Architected a self-hosted MDM platform for company-owned tablets with three parts: a TypeScript/Fastify API with 24 endpoints, an 8-screen Next.js dashboard with a live fleet map, and a Kotlin kiosk agent. It uses the Google Android Management API and Cloud Pub/Sub for QR zero-touch enrollment, kiosk lockdown and remote lock, reboot and wipe.
- Built an offline-first GPS telemetry pipeline: Room + WorkManager batch uploads → async ingest → Redis/BullMQ worker → TimescaleDB with compression, retention and object-storage archiving. In a production load test, 50 simulated devices sent 150k points with **zero errors** at **~46k points/s** ingest.
- Enforced privacy and security by design: work-hours-only location collection, an append-only audit log protected by database triggers, JWT sessions with rotating refresh tokens and reuse detection, Google-signed OIDC webhook verification and per-route Nginx rate limits.
- Brought production infrastructure live within a 5-day sprint. The GitHub Actions CI runs 35 tests, 12 automated compliance checks and secret scanning, backed by daily backups with a weekly automated restore test and a written runbook.

Forward Deployed Engineer

Nov 2024 – Present

Self-Employed · Freelance · Remote

Embedded with small businesses and independent professionals that have no engineering staff, turning their day-to-day workflows into production software. I deliver every project solo, from requirements through post-launch support.

Barbearia Tech — White-Label Booking Platform · Next.js 16 · React 19 · Neon Postgres · Vercel

- Built a config-driven white-label architecture: one brand file controls branding, locations, services, staff and payment methods, so onboarding a new barbershop takes a config change instead of new code.
- Delivered a 4-step booking flow backed by server-side availability per barber and timezone-aware scheduling. A role-scoped staff dashboard shows revenue, average ticket and cancellations, behind HMAC-signed sessions. The payment layer is provider-agnostic (Pix and card), so a real payment provider can be swapped in.

Lucky Clover — Multi-Role Sales & Payments Platform · React · Vite · Supabase · Deno Edge Functions · Mercado Pago

- Delivered a lottery-pool sales system as a TypeScript monorepo with 3 apps: customer, reseller point of sale and admin back office. It runs on Supabase with 47 migrations, 29 role-based RLS policies and 4 Edge Functions. Other pieces: Pix payments with HMAC-verified idempotent webhooks, scheduled pg_cron sync against an official results API, PDF/Excel reporting, and 58mm thermal-ticket printing on Android POS terminals.
- Led a security-hardening pass that closed a privilege-escalation path and a payment-bypass vector. Added per-user session revocation and fixed an XSS flaw in ticket printing.
- Published a sanitized public edition: stripped all secrets and personal data, and replaced the backend with an interface-compatible mock without touching any UI code.

Structural Engineering Portfolio · Astro · Tailwind CSS · TypeScript

- Designed and built a statically generated portfolio and lead-generation site for a BIM and structural-design engineer.

SELECTED PROJECTS

Mana Realm — Cross-Platform Desktop & Browser Game

Oct 2025 – Present

JavaScript · HTML5 Canvas · NW.js · Steamworks · GitHub Actions · Vercel · manarealm.vercel.app

- Ships a single vanilla-JS codebase to the web and to 4 desktop targets (Windows, Linux, macOS arm64/x64). A GitHub Actions native-runner build matrix produces the desktop builds. Steam integration runs in a separate helper process, which works around a crash when the desktop runtime loads native modules.
- Profiled rendering with the Chrome DevTools Protocol and cut GPU fill by **60%**, texture memory by **63%** and HUD raster time from 11.9 to 4.5 ms. Other optimizations: a zero-allocation frame loop, spatial hashing and adaptive quality tiers.
- Wrote a custom verification suite of ~500 checks that guards balance data, asset integrity and localization in 6 languages.

EDUCATION

Associate Degree — Systems Analysis and Development · Senac · 2024

LANGUAGES

English (Professional working proficiency) · Portuguese (Native)